

Preference Post-Training Qwen2.5-0.5B

Supervised Fine-Tuning, Reward Modeling, PPO, and Auditable Evaluation

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Abstract

This work studies a complete reinforcement learning from human feedback (RLHF) pipeline on QWEN2.5-0.5B-INSTRUCT with NVIDIA’s HELPSTEER3-PREFERENCE. Although the starting model is already instruction-tuned, one epoch of response-only supervised fine-tuning (SFT) is used as domain and preference adaptation and as the common reference policy for later alignment. A two-epoch scalar reward model and a KL-regularized Proximal Policy Optimization (PPO) policy are then trained with long-context, EOS-safe preprocessing and implementation details drawn from the N+ RLHF reproduction.

SFT reaches 72.02% teacher-forced completion-token accuracy on 1,917 retained validation pairs. The reward model ranks the human-preferred response correctly in 65.62% of those pairs. On a separate generation suite covering all 2,017 validation prompts, the learned reward model prefers PPO over the original instruction model in 1,027 cases, prefers Base in 981, and assigns 9 ties (50.92% PPO win rate). This narrow proxy-judge gain is accompanied by substantially greater length, cap hits, and repetition. The result is therefore presented as an inspectable alignment case study, not as evidence that PPO universally improves the base model. Complete outputs and a balanced curated subset are published for direct review.

1 Introduction

RLHF attempts to connect next-token language modeling with human judgments of helpfulness, relevance, correctness, clarity, and safety. Its canonical pipeline first adapts a policy to preferred demonstrations, learns a scalar reward from response comparisons, and then optimizes the policy under a constraint to its supervised reference [1, 2].

This project asks whether that pipeline can be implemented, measured, and audited on a half-billion-parameter model using rented notebook compute. The small model makes iteration accessible, but it also exposes capacity limits and makes reward-model error consequential. The Qwen2.5 model family is described in [7]. The work began with custom SFT, reward-model, and token-level PPO loops, then migrated the trainer mechanics to Hugging Face TRL while retaining repository-owned data contracts, model initialization, checkpoint records, evaluation, and qualitative audit.

Why SFT an instruction model? The starting checkpoint is already instruction-tuned, so this stage is not instruction tuning from scratch. It is continued SFT on the preferred HELPSTEER3-PREFERENCE responses.

This is justified for two reasons: it adapts the response distribution to the exact preference data used downstream, and it creates a reproducible reference policy shared by reward modeling, PPO, and the planned DPO comparison. Using the original instruction checkpoint directly would remove this controlled reference and make downstream differences harder to attribute. Continued supervised adaptation before preference optimization is also consistent with modern post-training pipelines [2].

The principal contribution is not a state-of-the-art model. It is an end-to-end study in which training and failure analysis receive equal weight: full responses, reward scores, length, EOS behavior, cap hits, repeated 4-grams, deterministic triage, and curated examples remain available for inspection.

2 Method

2.1 Data and preprocessing

HELPSTEER3-PREFERENCE contains 40,476 pairwise records over general, code, STEM, and multilingual tasks [4]. Each record contains one conversational context, two candidate responses, an overall preference in $\{-3, \dots, 3\}$, and up to three individual annotator judgments. Negative overall values favor response 1, positive values favor response 2, and zero is a tie.

Tied or invalid pairs are excluded from preference training, leaving 36,264 training and 1,917 validation records. All stages use the Qwen chat template and distinct PAD and EOS tokens. Completions are preserved through EOS; when a sequence exceeds 4,096 tokens, old prompt turns are removed before a final prompt-side token fallback. This reduces observed SFT training truncation from 38.47% at a 1,024-token budget to 0.83% at 4,096.

2.2 Supervised fine-tuning

For prompt x , preferred response $y = (y_1, \dots, y_T)$, and completion mask m_t , response-only SFT minimizes

$$\mathcal{L}_{\text{SFT}}(\theta) = - \sum_{t=1}^T m_t \log \pi_{\theta}(y_t \mid x, y_{<t}). \quad (1)$$

Prompt and padding tokens do not contribute to the loss. LoRA [5] rank-16 adapters are trained in all attention and feed-forward projections; the backbone remains frozen. The merged result supplies both the initial policy and the fixed reference for preference optimization.

2.3 Reward modeling

The reward model begins from the merged SFT backbone and adds a scalar head. For preferred y^+ and rejected y^-

under the same prompt, it minimizes the Bradley–Terry loss

$$\mathcal{L}_{\text{RM}}(\phi) = -\log \sigma(r_\phi(x, y^+) - r_\phi(x, y^-)). \quad (2)$$

Only score differences are identifiable; the sign of one isolated reward is not a quality boundary. A fixed offset estimated on preferred demonstrations is therefore subtracted consistently during PPO and evaluation.

2.4 KL-regularized PPO

PPO [6] receives a terminal reward-model score and token-level shaping against the frozen SFT policy:

$$r_t^{\text{KL}} = -\beta [\log \pi_\theta(y_t | s_t) - \log \pi_{\text{ref}}(y_t | s_t)], \quad (3)$$

$$R(x, y) = r_\phi(x, y) + \sum_t r_t^{\text{KL}}. \quad (4)$$

With generalized advantages $\hat{A}_t$ and $\rho_t = \pi_\theta(y_t | s_t) / \pi_{\text{old}}(y_t | s_t)$, the policy loss is

$$\mathcal{L}_{\text{PPO}} = -\mathbb{E}_t \left[\min\{\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t\} \right]. \quad (5)$$

The final path follows high-impact N+ implementation details [3]: zero policy dropout, behavior log probabilities matched to sampling temperature, a frozen SFT reference, an SFT-initialized reward model, controlled scalar-head initialization, reward centering, an RM-initialized critic, masked reward/advantage whitening, Adam $\epsilon = 10^{-5}$, and fixed-length sampling followed by EOS truncation. Responses without EOS receive reward -1 .

3 Training Results

3.1 SFT assessment

The one-epoch SFT run is stable. Validation loss changes only from 1.1183 at step 250 to 1.1127 at step 1,134, while completion-token accuracy remains near 72%. There is no late validation reversal suggesting material overfitting. The sequence budget retains nearly all responses, and the effective batch, learning rate, LoRA coverage, precision, and zero-dropout setting are conservative for continued adaptation of a 0.5B model.

These metrics do not prove better open-ended responses, but they establish the purpose required here: a stable HelpSteer-adapted policy and common reference. Accordingly, this checkpoint is frozen for subsequent experiments rather than rerun merely to obtain a lower training loss.

3.2 Reward model

The two-epoch reward model reaches 65.62% overall pairwise accuracy. Accuracy increases with preference strength: 61.23%, 66.33%, and 71.10% for strengths 1, 2, and 3. Domain accuracy is uneven: 70.23% on code, 62.91% on general, 59.26% on STEM, and 71.82% on multilingual examples. This is usable as a noisy preference signal, but not sufficiently reliable to serve as

ground truth, especially for correctness-sensitive STEM outputs.

An earlier custom reward model obtained 71.62% validation accuracy yet participated in a worse downstream PPO run. This is not evidence that lower accuracy is desirable; it shows that aggregate in-distribution rank accuracy does not capture reward scale, critic initialization, EOS handling, policy optimization, or behavior on policy-generated text.

3.3 PPO dynamics

The selected PPO run was configured for 12,000 episodes and stopped for full evaluation after 6,400 episodes and 100 optimizer updates. Average response-level objective KL was 1.8278 and the final value was 2.1648. This statistic aggregates over a response and is not directly comparable to per-token KL targets. Old-to-new update KL remained much smaller (mean 0.000640), the mean clip fraction was 0.00557, value loss fell from 25.32 to 5.50, and no empty-response collapse occurred. Optimization was numerically stable, but stability alone did not prevent reward exploitation.

4 Evaluation

4.1 Protocol

Base, SFT, and PPO each generate one response for all 2,017 validation prompts using the same tokenizer, prompt formatting, 3,072-token prompt limit, 1,024-token response cap, temperature 0.7, and top- $p = 1$. The same epoch-two reward model scores all prompt-response pairs. Pairwise tables derive from one shared response table, preventing comparisons from using different samples. Because the PPO training judge is also the evaluator, learned reward is treated as a diagnostic rather than independent human preference.

PPO narrowly crosses 50% against Base under the learned reward model, with the aggregate advantage concentrated in general prompts. It remains below Base on code and slightly below on STEM and multilingual prompts. The median PPO-minus-Base reward delta is only +0.0254, whereas the mean is +0.6497; a positive tail, including some flawed high-reward responses, raises the mean.

4.2 Stopping, repetition, and qualitative audit

PPO is substantially longer and more likely to exhaust the generation budget. The fraction of responses whose repeated word-level 4-gram mass exceeds 25% is 10.11% for Base, 20.08% for SFT, and 31.88% for PPO. At the stricter 50% threshold, the rates are 3.02%, 8.48%, and 15.82%. Thus, the learned-reward gain comes with an observable degradation in stopping and repetition.

Every evaluation row is assigned a deterministic triage label from reward margins, EOS/cap status, and repeated 4-grams. The rules identify 8 likely genuine PPO wins, 354 modest clean wins, 924 cases needing manual review, 228 repetition risks, 151 reward-model

Table 1: Final training profile. “Token accuracy” is teacher-forced next-token accuracy over preferred completion tokens, not task accuracy.

Stage	Data and budget	Adaptation	Optimization	Outcome
SFT	36,264 train; 1,917 val.; 4,096 total tokens	LoRA $r = 16$, $\alpha = 32$, dropout 0	1 epoch; effective batch 32; LR 5×10^{-6} ; cosine, 3% warmup	Train loss 1.0556; val. loss 1.1127; token accuracy 72.02%
Reward model	36,264 pairs; 1,917 val. pairs; 4,096 tokens	SFT backbone; scalar head; LoRA $r = 32$, $\alpha = 64$	2 epochs; effective batch 64; LR 5×10^{-6} ; centering coefficient 0.01	Val. loss 0.6166; pairwise accuracy 65.62%; mean margin 0.3578
PPO	24,018 unique prompts; 768-token rollouts	SFT policy/reference; RM-initialized value model	64 episodes/update; 4 PPO epochs; LR 3×10^{-6} ; KL coefficient 0.07	6,400 episodes; 100 updates; objective KL 2.1648 at final update

Table 2: Full policy-suite evaluation on 2,017 prompts.

Policy	Three-way wins	Win rate	Mean reward	Mean tokens	Median tokens	Cap hit
Base	718	35.60%	0.0803	398.0	331	8.82%
SFT	508	25.19%	0.0652	436.5	371	13.39%
PPO	775	38.42%	0.7300	577.8	520	27.42%
Tie	16	0.79%	—	—	—	—

false-positive risks, 288 severe repetition failures, and 64 strong PPO regressions. These labels prioritize review; they are neither human judgments nor LLM-as-judge verdicts.

The public artifact contains all 2,017 full prompts and responses, with secret-shaped strings redacted. A balanced 100-example subset contains 50 positive and 50 negative cases across all domains. Inspection finds real local gains, such as better multi-part coverage and compact recovery from meandering answers, alongside repeated loops, irrelevant continuation, malformed code, fabricated citations, and incorrect scientific procedures. Publishing both sides prevents the proxy score from becoming the entire account of the model.

5 Discussion

The strongest supported result is a systems-and-evaluation result. The project connects long-context preprocessing, parameter-efficient SFT, pairwise reward modeling, online PPO, checkpoint-safe execution, full-set generation, and transparent audit on a diverse preference dataset. PPO changes behavior and produces useful local gains, but the same run demonstrates how a numerically stable policy can exploit a limited scalar judge.

The central limitations are:

1. the reward model is both training signal and final quantitative judge;
2. no blinded, multi-rater human preference study is reported;
3. one stochastic generation per policy does not measure decoding variance;
4. response length confounds reward, completeness, and repetition;
5. a 0.5B policy and reward model have limited factual and long-context capacity; and

6. the selected PPO checkpoint was practical rather than chosen by an independent human metric.

5.1 Controlled DPO extension

The next experiment replaces reward-model optimization with Direct Preference Optimization (DPO) [8] while keeping the SFT checkpoint, preference records, tokenizer, 4,096-token budget, LoRA targets, and evaluation prompts fixed:

$$\Delta_{\theta} = \log \frac{\pi_{\theta}(y^{+} | x)}{\pi_{\text{ref}}(y^{+} | x)} - \log \frac{\pi_{\theta}(y^{-} | x)}{\pi_{\text{ref}}(y^{-} | x)}, \quad (6)$$

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E} \log \sigma(\beta \Delta_{\theta}). \quad (7)$$

The primary design uses standard sigmoid DPO with $\beta = 0.1$, one epoch, effective batch 32, LoRA rank 16, zero dropout, and precomputed reference log probabilities. Each non-tied pair appears once. HelpSteer3 preference strength is retained for stratified evaluation but is not silently interpreted as a linear utility weight; an explicit strength-weighted sensitivity run can be reported separately. The comparison will test whether direct offline preference learning avoids the reward hacking and rollout cost observed under PPO at this scale.

Further work should add blinded human review, repeated sampled generations and paired confidence intervals, reward-model hard negatives drawn from observed PPO failures, stronger correctness checks for code and STEM, and multi-metric checkpoint selection. Scaling the policy is justified only after the evaluator is less exploitable.

6 Conclusion

One-epoch continued SFT provides a stable HelpSteer-adapted reference, the reward model learns a meaningful but imperfect ranking signal, and PPO produces the

Table 3: Pairwise learned-reward results and PPO–Base domain breakdown. Win rates include ties in the denominator.

Comparison	Left wins	Right wins	Ties	Right win rate	Mean right–left delta
Base vs SFT	1,158	837	22	41.50%	−0.0151
Base vs PPO	981	1,027	9	50.92%	+0.6497
SFT vs PPO	840	1,164	13	57.71%	+0.6648
PPO vs Base by domain	Base wins	PPO wins	Ties	PPO win rate	Prompts
Code	250	188	0	42.92%	438
General	399	529	3	56.82%	931
STEM	126	118	1	48.16%	245
Multilingual	206	192	5	47.64%	403

strongest learned-reward result in the project. Yet the same evaluation reveals longer, frequently capped, and substantially more repetitive responses. The appropriate conclusion is neither that PPO failed nor that it solved alignment. The project shows that a small-model RLHF stack can be built and investigated rigorously, and that credible reporting requires training metrics, full responses, and failure analysis together. DPO now provides the cleanest controlled next comparison.

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